MATH 6660 — Final Project Proposal

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Title. Cohesive Zone Elements: Camanho's Bilinear Law, its Finite Element Implementation, and its Engineering Applications

Article. P. P. Camanho, C. G. Dávila, and M. F. de Moura, "Numerical simulation of mixed-mode progressive delamination in composite materials," *Journal of Composite Materials* **37**(16) (2003), 1415–1438. doi:10.1177/0021998303034505

Proposed project. Camanho, Dávila and de Moura model interface fracture with zero-thickness decohesion elements governed by a bilinear mixed-mode traction–separation law: a quadratic stress criterion for damage onset, the Benzeggagh–Kenane criterion for propagation, and a single irreversible damage variable [1, 2]. This project will derive the finite element formulation, in which the displacement may jump across the fracture surface Γ_c and the weak form gains an interface term, and show how the damage evolves under different loading conditions (pure opening, pure shear and compression–shear) using single-element simulations in MOOSE [3], an open-source finite element framework that implements this law. Finally, the project will apply the model to example problems, which may include composite materials, nuclear reactor components and natural fractures in subsurface rock.

References

[1] P. P. Camanho, C. G. Dávila, M. F. de Moura, Numerical simulation of mixed-mode progressive delamination in composite materials, *J. Compos. Mater.* 37 (2003) 1415–1438.

[2] P. P. Camanho, C. G. Dávila, Mixed-mode decohesion finite elements for the simulation of delamination in composite materials, NASA/TM-2002-211737, NASA Langley Research Center, 2002.

[3] G. Giudicelli et al., 3.0 – MOOSE: Enabling massively parallel multiphysics simulations, *SoftwareX* 26 (2024) 101690.